INCH-POUND

MIL-S-43357E
28 September 1990
SUPERSEDING
MIL-S-43357D
23 November 1983

MILITARY SPECIFICATION

SHIRT, SLEEPING, HEAT RETENTIVE AND MOISTURE RESISTANT, NYLON/ACETATE, TRICOT KNIT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of heat retentive and moisture resistant nylon/acetate, tricot knit sleeping shirt.

1.2 Classification. The shirts shall be of one type in the following sizes as specified (see 6.2).

Schedule of sizes

XX-Small	Small	X-Large
X-Small	Medium	XX-Large
	Large	

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
V-F-106	-	Fasteners, Slide, Interlocking
V-T-285	-	Thread, Polyester
DDD-L-20	-	Label: For Clothing, Equipage and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	-	Loads, Units: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43358	-	Cloth, Knitted, Nylon/Acetate, Tricot, Camouflage 483

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets
MIL-STD-1491	-	Glossary of Knitting Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF THE ARMY FIELD MANUAL

FM 10-280 - Mobile Field Laundry, Clothing Exchange, and Bath Operations

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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U.S. Army Natick Research, Development, and Engineering Center

Natick Formula IV - Laundering Procedure

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

COLOR ASSOCIATION OF THE UNITED STATES

Department of Defense Standard Shades Color Card for Sewing Threads

(Color cards may be available from the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Type number 304 - Stainless and Heat Resistant Steel
Type number 316 - Stainless and Heat Resistant Steel

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from the document may appear in the sample, in which case the document shall govern.

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3.1.2 Standard sample. The finished shirt shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.1.3 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Basic material. The cloth for the body, sleeves, and collar shall be Camouflage Green 483, conforming to MIL-C-43358.

3.2.2 Cuff material. The cloth for the cuffs shall be a true 1 x 1 rib, knit from 2 ply, 100 denier, multifilament, semi-dull, normal tenacity nylon yarn. The cloth shall conform to the physical requirements listed in table I when tested as specified in 4.4.1.

TABLE I. Physical requirements

Weight (oz./sq. yd.)	6.5 (min.)
Yarns per inch:	
Wales	23 (min.)
Courses	42 (min.)
Bursting strength (lbs.)	150 (min.)

3.2.2.1 Color. The color of the cloth for the cuffs shall be Camouflage Green 483.

3.2.2.1.1 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.2.2.1.2 Colorfastness. The dyed cloth shall show colorfastness to laundering, perspiration and crocking equal to or better than the standard sample (see 6.4). When no standard sample is available, the dyed cloth shall show "good" fastness to laundering and perspiration and shall show a Munsell Value for crocking not lower than 8.5. Testing shall be as specified in 4.4.1.

3.2.2.2 Finish. The dyed cloth shall be thoroughly scoured, and finished to meet the physical requirements in table I and the shrinkage requirements in 3.2.2.3.

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3.2.2.3 Shrinkage. The finished cloth shall not shrink or elongate more than 3.0 percent in length and 4.0 percent in width. Testing shall be as specified in 4.4.1.

3.2.3 Thread, polyester. The polyester thread for seaming and stitching shall conform to type I, class 1, size A, AA and B of V-T-285. The thread shall be dyed to match Olive Drab S-1, C.A. 66022 and shall show fastness to laundering equal to or better than the standard sample (see 6.4). When no standard sample is available, the dyed gimp shall show "good" fastness to laundering. Lubrication of the polyester thread by any means in the end item fabricators' plant prior to or during sewing is prohibited.

3.3 Combination size, identification and instruction label. Each shirt shall have a combination size, identification and instruction label conforming to type VI, class 14, of DDD-L-20. Label color shall be Medium Green, cable No. 70034, 70130, or 70131. The fastness to laundering requirements shall apply. The instructions shall be as follows:

a. Fixed Laundry: The sleeping shirt shall be laundered utilizing "Natick Formula IV". NOTE: DO NOT STARCH, BLEACH, OR PRESS.

b. Field Laundry: The sleeping shirt shall be laundered utilizing "Formula II" of FM 10-280. NOTE: DO NOT STARCH, BLEACH, OR PRESS.

c. Drying: The sleeping shirt shall be dried in a tumble-dryer, loaded not more than 2/3 of rated capacity at medium heat cycle (do not exceed 120° F). Remove immediately from dryer. Avoid over-drying.

3.4 Slide fastener. The slide fastener shall be type I, style 3, size LS; short tab pull, automatic locking, green polyester or green nylon monofilament in a continuous coil or ladder configuration conforming to V-F-106. The slide fastener shall be of sufficient length to accommodate the slide fastener opening indicated by the pattern marks. The tape extension at the top and bottom of the slide fastener shall be not less than 1 inch. The color of the slide fastener tape shall be Olive Green 507 and the color of all other slide fastener components shall approximate the shade of the tape.

3.5 Design. The shirt shall be a long sleeve, pullover type with a collar that converts to a turtleneck when the front opening slide fastener is closed to the top of the collar (see figure 1).

3.6 Patterns. Standard patterns which provide an allowance of 3/8 inch for collar and front and 1/4 inch for all other seams will be furnished by the Government. The Government patterns shall not be altered in any way and are to be used only as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns.

3.6.1 Pattern parts. Components parts shall be cut from the material specified in table II in accordance with the pattern parts indicated.

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TABLE II. List of pattern parts

Material	Nomenclature of parts	Cut parts
Nylon/acetate tricot knit cloth	Front	1
	Back	1
	Sleeves	2
	Collar	2
Nylon rib knit cloth	Cuffs	2

3.7 Construction. The construction shall conform in all respects to the requirements specified in table III and herein. Figure 1 is furnished solely for guidance and information. Should variation from the document appear in figure 1, the document shall govern.

3.7.1 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table III. Seam allowance shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the chain or underside of the stitch shall be on the inside of the shirts.

3.7.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewed.

3.7.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during end item inspection shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

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1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.7.1.2 Types 504, 505, 606 and 607 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.7.1.1.1a. and 3.7.1.1.1b.

3.7.1.3 Bartacking. Unless otherwise specified, bartacks shall be 3/8 ± 1/16 inch long and 1/8 ± 1/32 inch wide, and shall contain 28 stitches. Bartacks shall be free from thread breaks and loose stitching.

3.7.1.4 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met, and at least three or more tying, overlapping, or back stitches are used to secure the ends of stitching.

3.7.1.5 Thread ends. All thread ends that are visible on the finished item shall be trimmed to a length of not more than 1/4 inch unless otherwise specified.

3.7.2 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the shirt.

3.8 Manufacturing operations requirements. The shirt shall be made by and with the use of all operations listed in table III. Unless otherwise specified in table III, the contractor is not required to follow the exact sequence of operations. Any additional basting or holding stitching used to facilitate manufacture is permissible provided that the thread is removed or does not show on the finished shirt.

3.9 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table III, providing that the seam and stitch type are as specified and the finished component conforms to the required configuration. When a government finished shaper pattern is forwarded, the component shall conform to that pattern.

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NO.	TABLE III MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cut shirts. a. Cut the shirts in strict accordance with patterns furnished, which show size, directional lines, and notches for proper assembly of all parts. Parts shall be cut lengthwise in the direction of the wales. The napped size of the fabric shall be on the inside of the shirt. b. Cut all parts out of one piece of material.						
2.	Replacement of damaged parts. Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects and damages as specified in 4.4.2 are excluded and replaced with non-defective and properly matched material.						
3.	Shade and size marking. Mark, ticket or bundle all parts of the shirt to insure a uniform shade and proper assembly throughout the shirt. Any method of marking may be used except: a. Corrosive metal fastening devices. b. Sew-on tickets. c. Adhesive type tickets which leave an inpression or traces of paper or adhesive on the material after removal of the ticket.						

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1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

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NO.	TABLE III. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Join cuffs to sleeves.						
	Seam cuffs to sleeves.	606 or 607	FSa-1	12-16	AA	A	A
5.	Join shoulders.						
	Seam shoulders.	606 or 607	FSa-1	12-16	AA	A	A
6.	Seam sleeves to body.						
	Seam sleeves to armholes.	606 or 607	FSa-1	12-16	AA	A	A
7.	Join side and underarm seams.						
	Join side seams and underarm seams in a continuous operation.	606 or 607	FSa-1	12-16	AA	A	A
8.	Attach undercollar and label.						
	Stitch undercollar to neck of shirt from edge to edge with right sides together catching the label in the center.	301	SSa-1	8-12	B	B	B
9.	Attach fastener.						
	a. Overedge raw edge of front opening.	504 or 505	SSa-1	8-12	B	B	B
	b. Attach fastener to shirt opening and undercollar with top fastener stop 1/4 inch from top collar seam and tabs pulled in.	301	SSa-1	8-12	B	B	B

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TABLE III. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Finish collar.						
	a. Lay top collar on undercollar right sides together and stitch three sides stopping short of seam allowance on bottom edge.	301	SSa-1	8-12	B	B	B
	b. Turn collar, forcing out points and edges.						
	c. Turn in seam allowance of top collar and stitch to neck 1/16 to 3/32 inch from edge through all plies and continuing around collar and fastener for the raise stitching. A 1/4 inch tack is required at beginning and end.	301	LScf-2(b)	8-12	B	B	B
11.	Hem shirt.						
	Turn up bottom of shirt and seam, making 1 ± 1/4 inch hem.	504	EF1-1	8-12	A	A	A
	-or-						
	Turn in bottom of shirt 1 inch with the raw edge turned under 1/4 inch and stitch 1/16 to 1/8 inch from turned in edge. The hem shall finish 3/4 ± 1/4 inch wide.	301	EFb-1	8-12	A	A	
12.	Clean shirt.						
	Trim all ends of stitching and remove loose threads, inside and outside of shirt. Remove all spots and stains.						

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TABLE III. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
13.	Tack cuffs. The stitching chain at bottom of cuff shall be turned under 1/2 to 5/8 inch and tacked, beginning 1/8 to 3/16 inch from bottom edge of cuff, with 1/2 ± 1/16 inch bartack or straight line tack.	Bartack or 3 line tack		18 per or 8-12	B	B	

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3.10 Finished measurements. The measurements of the finished shirt shall conform to the requirements specified in table IV (see figure 1). All measurements shall be taken with the slide fastener closed to bottom of the collar and the shirt placed smooth and flat on a table.

TABLE IV. Finished measurements (in inches)

		XX-Small	X-Small	Small	Medium	Large	X-Large	XX-Large	Tolerance (+ or -)
A	Half shirt width 1/	17-3/4	19-3/4	21-3/4	23-3/4	25-3/4	27-3/4	29-3/4	3/4
B	Overall length	28	29	30	31	32	33	34	3/4
C	Sleeve length 2/	21-1/2	22	22-1/2	23	23-1/2	24	24-1/2	1/2
D	Zipper length	12-1/4	12-1/4	13-1/4	13-1/4	13-7/8	13-7/8	13-7/8	1/4
E	Half sleeve width 3/ taken 9-1/2 inches below shoulder seam	6-3/4	7-1/8	7-5/8	8-1/8	8-5/8	9-1/4	9-7/8	1/4
F	Collar height	4-3/8	4-3/8	4-3/8	4-3/8	4-3/8	4-3/8	4-3/8	1/4
G	Cuff length 3/	4-1/2	4-1/2	4-1/2	4-1/2	4-1/2	4-1/2	4-1/2	1/4
H	Cuff width	3-1/2	3-1/2	3-1/2	3-1/2	3-1/2	3-1/2	3-1/2	1/4
I	Hem width	1	1	1	1	1	1	1	1/8

1/ Across width of body, 1 inch below armhole seam.

2/ Measurement to be taken from intersection of shoulder seam and armhole seam to the top of the cuff with the sleeve laying smooth and flat.

3/ Paired component measurements shall not be uneven by more than 1/2 inch.

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3.11 Workmanship. The finished shirt shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assur supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1.3 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Component testing. In addition to testing provisions contained in subsidiary documents, testing shall be performed on the components listed in table V for the characteristics noted. Unless otherwise indicated, testing shall be conducted in accordance with FED-STD-191. All test reports shall contain the individual values used in expressing the final result. The lot shall be unac-ceptable if one or more sample units fail to meet any of the test requirements specified. The sample unit for testing the cuff rib material shall be 3 yards, full width, and the lots shall be expressed as yards. The lot size and sample size shall be in accordance with the following:

Lot size	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE V. Component tests

Component	Characteristic	Requirement paragraph	Test method
Cuff rib knit material			
	Material identification	3.2.2	1/
	Type of knit	3.2.2	1/
	Ply	3.2.2	1/
	Denier	3.2.2	1/
	Multifilament	3.2.2	1/
	Luster	3.2.2	1/
	Tenacity	3.2.2	1/
	Weight (oz./sq. yd.)	Table I	5041
	Wales per inch	Table I	5070
	Courses per inch	Table I	5070
	Bursting strength	Table I	5120
	Colorfastness to:		
	Laundering	3.2.2.1.2	5614
	Perspiration	3.2.2.1.2	5680
	Crocking	3.2.2.1.2	5651
	Shrinkage	3.2.2.3	5556 2/

1/ Unless otherwise specified, a certificate of compliance is required and will be acceptable for the stated requirements.

2/ Wool laundering procedure with tumble dry not to exceed 140°F.

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4.4.2 End item visual examination. The end items shall be examined for the defects listed in table VI. The lot size shall be expressed in units of shirts. The sample unit shall be one shirt. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE VI. Visual defects

Examine	Defect	Classification Major	Minor
Material defects	Hole, cut or tear, smash, run, or dropped stitch 1/	101	
	Needle chew	102	
	Thin or uneven area	103	
	Mend 1/	104	
	Slub (more than twice the size of the normal diameter of the yarn), float or snag 1/		201
Color and shade	Color not as specified	105	
	Variation in shade within a part or between two or more parts 2/ 3/		202
	Streaks clearly visible 3/		203
	Barre 1/		204
Cleanness	Spot or stain up to 1/4 inch in largest dimension		205
	Spot or stain more than 1/4 inch in largest dimension	106	
	Two or more thread ends not trimmed to 1/2 inch or less in length		206
	One or more shade tickets not removed		207
	One or more size or shade markings clearly visible on outside 3/		208
Component and assembly	Any component part or required operation omitted or added (unless otherwise classified herein)	107	
	Any operation not performed as specified (unless otherwise classified herein)		209
	Napped side of tricot knit fabric not on inside of shirt	108	

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TABLE VI. Visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Cutting	Any component part not cut in accordance with directional lines as indicated on patterns or in accordance with specification requirements	109	
Seams and stitching	Any open seam (except on hems)	110	
	Open seam on hems:		
	- 1/4 to 1 inch		210
	- over 1 inch	111	
	NOTE: A seam shall be considered as being open when one or more stitches joining a seam are broken, or when two or more successive skipped stitches occur.		
	Seam or stitch type not as specified	112	
	Parts of shirt badly pleated, caught or twisted in any type of stitching	113	
	Seam twisted or puckered		211
	Thread breaks (all stitch types) backstitched less than 1/2 inch		212
	Ends of all seams and stitchings produced with 301 stitch type not caught in other seams or stitchings, or backtacked less than 1/4 inch		213
	Ends of a continuous line of stitching inch overlapping less than 1/2 inch		214
	Raw edges:		
	Any on outside	114	
	On inside 1/8 to 1/2 inch (inclusive)		215
	On inside, over 1/2 inch	115	
	Runoff, or any line of stitching not catching all plies of material to be seamed:		
	- up to 1/2 inch (inclusive)		216
	- more than 1/2 inch	116	

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TABLE VI. Visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams and stitching (cont'd)	Stitch tension:		
	Loose tension resulting in a loose seam	117	
	Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	118	
	One stitch per inch more or less than specified		217
	Two or more stitches per inch less than specified minimum or more than specified maximum	119	
Hems	Omitted	120	
Tack	Tack omitted or not of required length at both ends of raised stitching on collar and fastener	121	
Fastener	Puckers or pleats around fastener	122	
	Fastener top stop more than 1/4 inch from point of collar		218
	Fastener not working properly	123	
Label or markings	Missing, incorrect, illegible, misplaced, or not attached as specified	124	

1/ For knitted fabric imperfection defects, see MIL-STD-1491.

2/ All parts shall be examined with wales and courses running in same direction.

3/ At normal inspection distance (approximately 3 feet).

4.4.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table IV. The lot size shall be expressed in units of shirts. The sample unit shall be one shirt. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: Incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Content	Number of unit packs per shipping container is more or less than specified.

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each shirt shall be folded with fastener closed to bottom of collar to approximate dimensions of 14-1/2 by 11-1/2 inches and secured with stainless steel (AISI type 304) pins, tin or nickel-plated brass pins, aluminum clips or plastic fasteners. Each folded shirt shall be unit packed in a snug-fitting, flat style, clear polyethylene film bag conforming to A-A-50083.

5.1.2 Commercial. The shirts shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Thirty shirts of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s, of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, front up, collars alternated front to rear two in length, one in width, and fifteen in depth. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed and reinforced in accordance with method III, and waterproofed in accordance with method V of the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Thirty shirts of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 200 of PPP-B-636. Level A unit packs shall be packed flat, front up, collars alternated front to rear, two in length, one in width, and fifteen in depth. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall be closed in accordance with method II as specified in PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Shirts, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), shirts packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, and IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 3 in accordance with appendix of MIL-STD-147.

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5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The shirts are intended for use as a component of the hot weather sleeping gear and as a supplement to the cold weather system.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1.3, 4.3 and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant garde fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also in include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Subject term (key word) listing.

Garment
Sleepwear

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Navy - MC
Air Force - 82
DLA - CT

User activity:

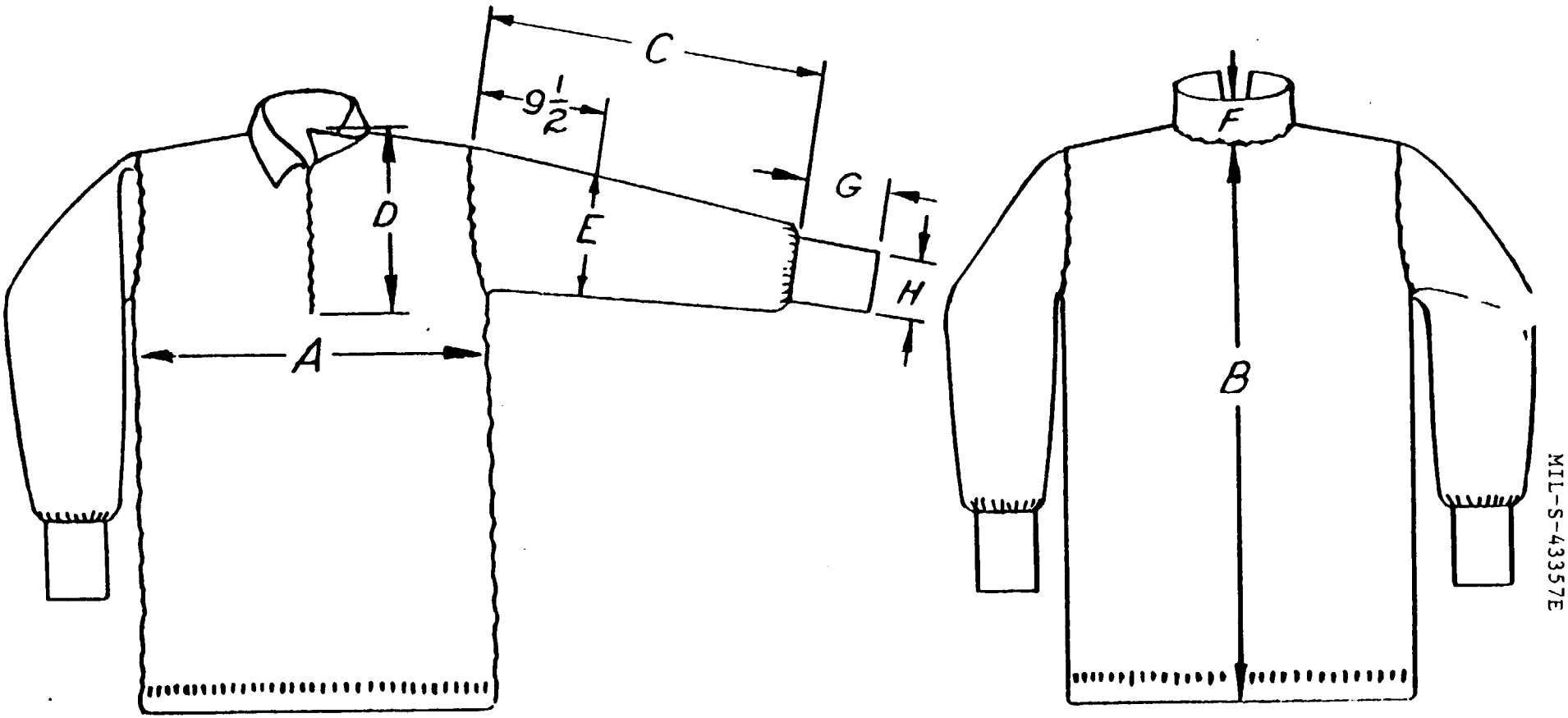
Air Force - 45

Preparing activity:

Army - GL

(Project 8415-0728)

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MIL-S-43357E

FIG 1, SHIRT, SLEEPING, HEAT RETENTIVE AND MOISTURE RESISTANT, NYLON/TRIACETATE TRICOT KNIT

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2-1-1604

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-43357E	1990 September 28

3. DOCUMENT TITLE
SHIRT, SLEEPING, HEAT RETENTIVE AND MOISTURE RESISTANT, NYLON/ACETATE, TRICOT KNIT

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-5221	(2) AUTOVON 256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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